



Biological and Microbial Control

Ethanol tolerance of fungal strains underlies biocontrol of invasive ambrosia beetles

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Entomopathogenic fungi are widely recognized as biological control agents showing potential against insect pests that are difficult to manage with chemical pesticides. Invasive ambrosia beetles threaten a wide variety of tree crops and prove challenging to manage with chemical pesticides due to their unique life history: entrenched in the xylem of tree hosts, feeding on symbiotic fungi. Many ambrosia beetle species attack trees that are flood-stressed and producing ethanol, a key component of growth for their fungal symbiont. Ethanol causes osmotic and chaotropic stress in microbes and could prevent successful biocontrol of ambrosia beetles using insect-killing microbes. To investigate the potential of entomopathogenic fungi to manage ambrosia beetles, we conducted a series of experiments using 3 species in eastern North America (*Xylosandrus germanus*, *Xylosandrus crassiusculus*, and *Anisandrus maiche*). We tested the efficacy of 3 microbial-based commercially available biopesticides (Botanigard 22 WP: *Beauveria bassiana*; Met52: *Metarhizium brunneum*; and PFR-97 20% WDG: *Cordyceps javanica*) in a series of environments with and without ethanol. We found that *B. bassiana* was the most effective against all 3 beetle species, but that its pathogenicity was heavily dependent on the ethanol-tolerance of the strain used. We isolated 2 strains of *B. bassiana* from separate batches of Botanigard that demonstrated significant differences in ethanol tolerance, as well as pathogenicity, radial colony growth, and spore production in the presence of ethanol. Our results emphasize the importance of pest natural history in control methods, while revealing that the performance of biocontrol products varies across microbial species and exhibits strain-dependency.

Keywords: invasive pests, entomopathogenic fungi, tree crops, environmental stress, obligate mutualism

Introduction

Biological control methods serve as an important alternative to chemical pesticides, especially when pests are poorly controlled with traditional pesticide applications. While many pest species are difficult to manage due to pesticide resistance, others escape pesticide control due to their life history and ecology (Castrillo et al. 2011, Carrillo et al. 2015, Shields and Testa 2017, Rim et al. 2019, Bras et al. 2022, Gao et al. 2022, Mhatre et al. 2022). For these pests, leveraging natural enemies (predators, parasitoids, and pathogens) can lead to novel, effective biological control methods (Lacey et al. 2015, Shields and Testa 2017, Wang et al. 2019, Reverchon et al. 2021). Unlike chemical pesticides, biological controls are living organisms

that will differ in their management effectiveness and environmental tolerances through genotypic and phenotypic variation (Cruz et al. 2006, Muirhead et al. 2012, Velivelli et al. 2014, Valero-Jiménez et al. 2016, Lommen et al. 2017, Bonaterra et al. 2022, Lartigue et al. 2022). For biological control to be successful, it is necessary to fully investigate the ecological interactions among all organisms (crop, pest, and control agent), as well as the surrounding environmental conditions.

Insect-killing (entomopathogenic) fungi are a promising avenue for biological control, which have co-evolved alongside insects to conquer the defensive mechanisms of their hosts (Humber 2008, Pell et al. 2010, Fernandes et al. 2012, Lacey et al. 2015, Mantzoukas and Eliopoulos 2020, Reverchon

et al. 2021, Saha et al. 2025). Entomopathogenic fungi lack the human, environmental, and non-target species health hazards of chemical pesticides (Goettel et al. 2001, Zimmermann 2007, Alabouvette and Cordier 2011, Koch et al. 2018, Huang et al. 2019) and can be a cost-effective alternative for growers (Bale et al. 2008, Barratt et al. 2018, Mantzoukas and Eliopoulos 2020). Several species are readily available as commercial biopesticides (including *Beauveria bassiana* [Bals.-Criv.], *Metarhizium brunneum* [Petch], and *Cordyceps javanica* [Bally] previously *Isaria fumosorosea* [Wize]). These fungi are widely used across a diverse range of insect pest species infesting multiple crop types (Zimmermann 2008, Mascarin and Jaronski 2016, Stone and Bidochka 2020). Entomopathogenic fungi are of particular interest for use against insect pests whose life history precludes successful management by chemical pesticides, which rely on contact or ingestion (Castrillo et al. 2011). Entomopathogenic fungi produce asexual conidia that adhere to the insect cuticle, providing a potential solution to pests that evade chemical pest management.

Invasive ambrosia beetles are pests of a variety of trees and shrubs that include native species and economically important crop species, including ornamental, nut, and fruit trees (Fraedrich et al. 2008, Eskalen et al. 2013, Na et al. 2018, Gugliuzzo et al. 2021, Ranger et al. 2021, Monterrosa et al. 2022). Adult beetles bore galleries into the sapwood of stressed trees where they cultivate mutualistic fungi that serve as food for themselves and their offspring (Weber and McPherson 1983, Ranger et al. 2016, 2021). Although the symbiotic fungus may, in rare cases, be pathogenic to the host tree, beetle infestations generally result in branch die-back, tree decline, and even tree death (Ranger et al. 2016, Agnello et al. 2021). We frame our research around apple orchards, a widely cultivated tree crop of New York State. In our study, we focus on 3 ambrosia beetles that are economically impactful in tree crops. *Xylosandrus germanus* (Blandford) (Coleoptera: Curculionidae: Scolytinae) (CABI 2019), the black stem borer, and its congener *Xylosandrus crassiusculus* (Motschulsky) (CABI 2021), the granulate ambrosia beetle, are native to Asia, but are now widespread in Europe and North America (Gomez et al. 2018). *Anisandrus maiche* (Kurentsov), a third east Asian species, is an emerging invasive pest species whose impact is relatively unknown and has currently been found infesting walnut and apple orchards (Rabaglia et al. 2009, 2019, Young et al. 2021, Thurston et al. 2022, Tobin and Ginzel 2023, Tobin et al. 2024).

All 3 of these ambrosia beetle species are obligate mutualists with species from a genus of Microascales fungi: the *Ambrosiella* (Harrington et al. 2014, Mayers et al. 2015, 2017). These beetle species are closely coevolved and usually stably associated with specific *Ambrosiella* fungi (Harrington et al. 2014, Mayers et al. 2015, 2017). Adult female beetles store and provide nutrients for fungal propagules in an organ known as a mycetangium (or mycangium) (Fraedrich et al. 2008, Harrington et al. 2014). During the development of a new gallery, overflowing propagules from the mycetangium inoculate the tunnel walls with the fungal symbiont (Mayers et al. 2017). The offspring of the founding female feed solely on their fungal symbiont before collecting propagules into their own mycetangia and dispersing to new trees (Mayers et al. 2015). The obligate nature of the mutualism between ambrosia beetles and their fungal symbiont makes it necessary to consider the intricacies of both the beetle and fungal life

history, development, and physiology when developing management strategies for these pests.

Ambrosia beetles are difficult to manage because they reside within their host for most of their life cycle (Castrillo et al. 2011). Most chemical pesticides rely on contact or ingestion to cause mortality; contact is challenging due to the minimal time ambrosia beetles spend outside of their galleries, and ingestion is nonviable since the beetles feed solely on their symbiotic fungi (Gerolt 1969, Bagnères et al. 2009). Management of *Xylosandrus* species has typically involved the application of insecticides such as permethrin to prevent beetle attack; repellents and host plant defense compounds have also been investigated (Ranger et al. 2016, Agnello et al. 2021, Gugliuzzo et al. 2021, Gugliuzzo et al. 2023b, Parajuli et al. 2023, Govindaraju and Joseph 2025). Although chemical treatments can reduce infestations, they do not prevent all ambrosia beetle colonization. Entomopathogenic fungi are a possible alternative management strategy. Fungal spores recognize their host on contact, germinate, and then penetrate the cuticle using mechanical and enzymatic pressure to ultimately proliferate in the host hemolymph until the host is overwhelmed (Lovett and St. Leger 2017). Fungal spores could be carried into the gallery with the female, possibly resulting in her death and/or disease transmission to her offspring. Previous studies have shown mixed lethal and sublethal effects of several entomopathogenic fungi species (including those used in this study) against various ambrosia beetles (Castrillo et al. 2011, Carrillo et al. 2015, Gugliuzzo et al. 2023b). Although entomopathogenic fungi may not fully prevent beetle colonization and gallery formation, due to delayed mortality of the foundress, they have the potential to reduce oviposition and even infect existing offspring in the galleries (Castrillo et al. 2011, 2013). Thus, entomopathogenic fungi can possibly both prevent and reduce damage caused by beetle attacks.

An important aspect of ambrosia beetle life history, and therefore its management with biological control agents, is that several invasive species preferentially infest stressed trees, especially trees weakened by flood stress or low temperature stress (Ranger et al. 2021, Gugliuzzo et al. 2023a). Ethanol is produced and emitted from stressed trees as a result of anaerobic respiration (Kimmerer and MacDonald 1987) and it serves multiple roles in the system: an attractant cue for dispersing beetles, a necessary nutritional element for its symbiotic fungus, and an environmental constraint on fungal competitors and pathogens (Ranger et al. 2013, 2018, 2021, Lehenberger et al. 2020, Cavaletto et al. 2023, Gugliuzzo et al. 2023a). Ethanol causes multiple types of cellular stress in microbes (Hallsworth 1998, Hallsworth et al. 1998, 2003, Dao and Dantigny 2011, Cray et al. 2015), which is potentially what allows the fungal symbiont of ambrosia beetles to dominate their galleries, avoiding competition and pathogens (Lehenberger et al. 2020). Since ethanol is necessary to the beetles for tree colonization, fungal growth, and brood development, ethanol must be considered in the assessment of biocontrol agents. If an entomopathogenic fungus is employed as a biological control against ambrosia beetles, its tolerance to ethanol stress will be a key component of success or failure in beetle management.

To determine if using entomopathogenic fungi against ambrosia beetles is a feasible control option, we conducted 3 experiments plus a follow-up fungal bioassay (Fig. 1). We first tested the efficacy of 3 commercial entomopathogenic species of fungi (Botanigard or *B. bassiana*, Met52 or *M. brunneum*,

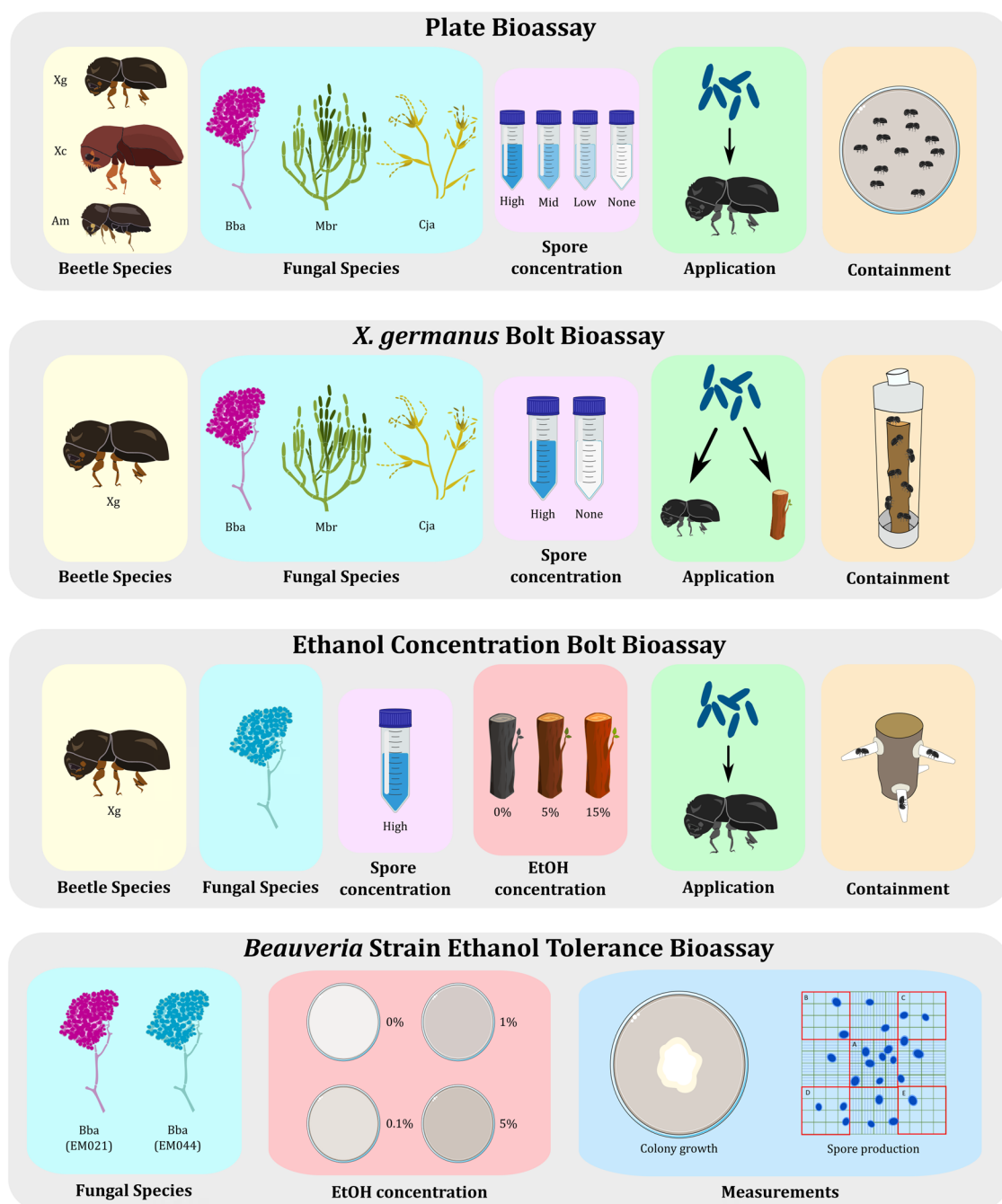


Fig. 1. Methods figure to provide an overview of the experimental design, involved organisms, and methods of the 4 bioassays conducted.

and PFR-97 or *C. javanica*) against 3 species of ambrosia beetle pests (*X. germanus*, *X. crassiusculus*, and *A. maike*) in the Plate Bioassay. From these results, we narrowed our focus to *X. germanus*, the most prevalent ambrosia beetle pest in New York apple orchards. In the *X. germanus* Bolt Bioassay, we assessed the potential of the 3 entomopathogenic fungal species against *X. germanus* in cut sections of beech trees (bolts) soaked in ethanol, to approximate the environment of a stressed tree where beetles establish galleries and produce offspring. Building from these results, we assessed the impacts of ethanol concentration in soaked sections of beech on the pathogenicity of *B. bassiana* against *X. germanus* in the Ethanol Concentration Bolt Bioassay. Our experiments revealed that

the pathogenicity of different strains of *B. bassiana* may be differentially affected by ethanol concentration. Therefore, we conducted ethanol tolerance bioassays on rich media plates in the *Beauveria* Strain Ethanol Tolerance Bioassay.

Methods

Ambrosia Beetles

Ambrosia beetles were collected in New York state using bottle traps baited with ethanol lures (95% ethanol, release rate 65 mg/d at 30 °C; ChemTica Internacional, S. A., Heredia, Costa Rica) during the summers of 2022 to 2024. All beetle species were reared through at least 1 lab generation before use

in experiments. The laboratory colonies were reared in either a saw-dust based artificial diet (adapted from [Castrillo et al. 2011](#)), or in 7 cm sections of beech bolts (*Fagus grandifolia* Ehrh.) soaked in a solution of 5% v/v ethanol and 5% v/v acetic acid. Beech trees of 2 to 5 cm in diameter were harvested in December 2023 and April 2024 and stored at 5 °C wrapped in plastic to prevent desiccation. For beetle rearing, bolts were soaked for 24 h in a solution of distilled water, 5% ethanol, and 5% acetic acid. Beetles were mass surface-sterilized before being added to bolt sections by rinsing in 0.01% Tween 80, 75% ethanol, and 2% bleach, followed by 2 washes of distilled water. Bolts were kept at room temperature (24 to 26 °C) until beetle emergence and use in experiments.

Entomopathogenic Fungi

We selected 3 species of widely used biocontrol fungi. Two species of entomopathogenic fungi were obtained as commercial products (*B. bassiana* conidia from Botanigard 22WP (Laverlam International Corp., Butte, MT) and *C. javanica* blastospores from PFR-97 20% WDG (Certis USA LLC, Columbia, MD)), and the third, *M. brunneum* (ARSEF #5198, formerly manufactured as Met52), was accessed from the Agricultural Research Service Entomopathogenic Fungi collection (ARSEF) ([Table 1](#)). Commercial products were prepared the day of use in distilled water at the highest concentration recommended by the manufacturer (3 to 5×10^7 cells/ μ L). In the Plate Bioassay, the commercial fungal products were also diluted to 1×10^5 cells/ μ L and 1×10^3 cells/ μ L to create a concentration gradient.

M. brunneum was cultured on potato dextrose agar (Difco Potato Dextrose Agar Mix; 0.4% potato starch, 2% dextrose, 1.5% agar; BD Biosciences, Sparks, MD) at least 10 d prior to experimental setup to allow for full maturation of spores. Spores were collected from plates into a spore solution ([Supplementary Materials](#)). A 12- μ L aliquot of spore suspension was placed on each side of a hemacytometer (Fisher Scientific, Pittsburgh, PA, USA) and counted to determine spore solution concentration. The spore solution was then diluted to 1×10^7 cells/ μ L (to approximate the highest concentration of the commercial products), and again to 1×10^5 cells/ μ L and 1×10^3 cells/ μ L for the Plate Bioassay.

Two strains of *B. bassiana* were used through the course of these studies. *B. bassiana* strains were cultivated from each batch of Botanigard by isolating the fungi from mycosed cadavers from the Plate Bioassay and the Ethanol Concentration Bioassay. We isolated *B. beauveria* from mycosed beetles as an efficient method of culturing infective strains without the adjuvants and potential contaminants from the commercial product. Strains EM021 (ARSEF #14872) and EM044 (ARSEF #14843) were isolated from Botanigard purchased in the spring of 2023 and spring 2024, respectively.

Plate Bioassay

To test the efficacy of our entomopathogenic fungi to kill ambrosia beetles at varying concentrations, we used *B. bassiana*, *M. brunneum*, and *C. javanica* against 3 ambrosia beetles species: *X. germanus*, *X. crassiusculus*, and *A. maiche*. Beetles ($n=20$ to 24) were directly treated with a high (1×10^7 cells/ μ L for *M. brunneum*; 3 to 5×10^7 dose for *B. bassiana* and *C. javanica*), medium (1×10^5 cells/ μ L), or low (1×10^3 cells/ μ L) dose of spores. Beetles were submerged in 20 mL of spore solution in a 50 mL tube, and vortexed for 30 s. Beetles were transferred to sterile petri dishes lined with sterile filter paper and allowed to dry for at least 5 min. Water (DI) was used in control groups. Treatments were replicated 4 times for each beetle species. Beetles were housed in parafilm-sealed 4% water agar plates at room temperature (24 to 26 °C) with a 0 L/24D light cycle. Plates were observed daily for mortality for 14 d. Beetle cadavers were removed from plates on the day of death and surface sterilized by washing in 70% ethanol for 10 s, followed by 2 washes in sterilized DI water. Cadavers were placed on 1.5% water agar plates and observed 5 d post death for signs of fungal outgrowth from the body (mycosis).

X. germanus Bolt Bioassay

To test the pathogenicity of our entomopathogenic fungi against *X. germanus* in a natural substrate, we set up 2 application treatment groups (directly treated beetles, and treated bolts) with 3 fungal treatment groups (*B. bassiana*, *M. brunneum*, and *C. javanica*) and a no-fungi water control for a 2 by 4 factorial experiment. All bolts were soaked in an ethanol solution to attract ambrosia beetles and allow for successful gallery formation; ethanol is a key cue for beetle digging and a necessary constituent for *Ambrosiella* growth. As such, we could not exclude ethanol and examine the effects of biological control fungi on life history such as gallery establishment and brood production. All fungi, regardless of application treatment, were applied at a 10^7 spore concentration. Beetles for each treatment group ($n=24$) were split between 2 bolts within the same application treatment to reduce gallery density and improve dissection resolution. Each application treatment and fungal treatment groups were replicated 4 times.

Beech bolts were cut to ~24 cm lengths and stored at 4 °C in plastic bags until use. Bolts were soaked in a 15% ethanol (v/v) solution for 72 h. After soaking, bolts were allowed to air dry until the surface was dry to the touch. Each bolt was given a unique ID, assigned to a treatment, and measured. In the beetle treatment group, beetles were vortexed in high concentration spore solutions before being transferred to untreated bolts (see methods in Plate Bioassay). In the treated bolts group, spore solutions were applied to bolt surfaces using a H series air brush (Passshe Airbrush company; Kenosha, WI 53144).

Table 1. Information on fungal species, strains, products, and manufacturers for all fungi used in experiments

Species	ARSEF Strain #	Product	Manufacturer	Experiments
<i>Beauveria bassiana</i>	14872	BotaniGard 22WP (EM021)	Laverlam International Corp.	Plate Assay, <i>X. germanus</i> Bolt Assay, <i>Beauveria</i> Strain Ethanol Tolerance
<i>B. bassiana</i>	14843	BotaniGard 22WP (EM044)	Laverlam International Corp	Ethanol Concentration Bolt Assay, <i>Beauveria</i> Strain Ethanol Tolerance
<i>Metarhizium brunneum</i>	5198	—	—	Plate Assay, <i>X. germanus</i> Bolt Assay
<i>Cordyceps javanica</i>	3682	PFR-97 20% WDG	Certis USA LLC	Plate Assay, <i>X. germanus</i> Bolt Assay

with 1.06 mm head and the compressor set to 40 psi. Twenty milliliters of each spore solution was loaded into a glass bottle (H-3-OZ 3 oz./88 cc Color Bottle Assembly, Passshe Airbrush company; Kenosha, WI 53144). The bolt was sprayed until runoff while rotating on a pottery wheel to ensure even coating on all surfaces (including cut ends). Bolts were allowed to surface dry before further handling. All equipment was sterilized with 70% ethanol and exposed to UV light in a closed biosafety cabinet between fungal treatments.

Beetles and bolts were housed in 30.5 cm long plastic tubes with a 7.6 cm diameter. The ends were sealed with rubber caps. Bolts were held in place inside the tube using semicircular pieces of dense foam, which prevented bolts from rolling and crushing beetles, and kept the cut end of the bolt from lying flat against the bottom cap. Twelve beetles were added to each bolt tube, as higher densities result in clustered galleries that cannot be differentiated upon dissection. Gallery entrances were marked 24 h after establishment, and given a unique ID. Beetles were left undisturbed on bolts for 14 d, after which bolts were cut into 2.5 to 7.5 cm sections, avoiding cutting through galleries by leaving ~1 cm of space on either side of the gallery entrance. These sections were dissected using hammers and wood chisels, and the galleries were examined to determine foundress mortality, brood presence and number, the physical structure of the gallery, and the physical appearance of the *Ambrosiella* fungi (Supplementary Fig. S1). Adult beetles that were found dead within the tube (ie. did not dig or exited the gallery after establishment) or galleries were set up for mycosis following the same protocol as in the Plate Bioassay.

Ethanol Concentration Bolt Bioassay

We investigated the effect of ethanol concentration in beech bolts on the pathogenicity of *B. bassiana* against *X. germanus* by directly treating all beetles (Fig. 1; see Plate Bioassay) with a 3 to 5×10^7 cells/ μ L dose of Botanigard (batch 2024, strain EM044). We had 4 treatment groups: (i) beech bolts soaked in 5% ethanol for 72 h, (ii) beech bolts soaked in 15% ethanol for 72 h, (iii) a negative control of beech bolts soaked in water for 72 h, and (iv) a positive control of beetles housed on a 4% water agar plate. We included a non-ethanol bolt control to account for other variables introduced by the wood substrate, without the expectation of beetle digging or gallery establishment. Each treatment consisted of 20 beetles, replicated 4 times. Each beetle ($n=20$ per treatment) was housed individually in a 1.5 mL micro-centrifuge tube attached to the bolt surface using dental wax. Four beetle arenas were attached to each 7 cm piece of beech bolt, evenly spaced across the surface, for a total of 5 bolt sections per treatment, per replicate. Bolts and beetles were housed in plastic containers (18.73 cm W $\times$ 7.62 cm H; Pioneer Plastics, KY) for 2 wk after beetle treatment and arena placement. The positive control treatment was established following the protocol in the Plate Bioassay. Beetles were checked daily for survival and digging status (if on bolts) for 14 d. At the termination of the experiment, bolts were dissected using a wood chisel and hammer, following the protocol described in *X. germanus* Bolt Bioassay, with the omission of cutting bolts into sections.

Beauveria Strain Ethanol Tolerance Bioassay

We investigated the ethanol tolerance of 2 strains (batch 2023, strain EM021, and batch 2024, strain EM044) by conducting growth bioassays on potato dextrose agar with 0%, 0.1%, 1%, or 5% ethanol by volume incorporated. These values were

chosen as a gradient between 0% and 5% ethanol used to treat bolts in the Ethanol Concentration Bolt Bioassay, to determine if small amounts of ethanol could elicit detrimental effects. Each treatment was replicated 5 times. Spore solutions were made by scraping small amounts of spores from 14-d-old cultures with a sterile pipette tip and transferring to sterile 0.01% Tween 80. Each strain was point-inoculated from the spore solution onto experimental plates using sterile pipette tips. Plates were photographed every 48 h for 8 d, for a total of 4 time points. Plate images were analyzed using the Fiji (ImageJ 1.54p) area tool to determine approximate radial growth at each time point. At the final time point, we quantified spore production for the 0% and 5% treatments for both *B. bassiana* strains. We did not analyze the spores from the 0.1% and 1% treatments because the radial growth did not significantly differ from that of the 0% ethanol treatment in either strain. Spores were quantified by scraping the surface of the colonies with a sterilized spatula and transferring spores to 5 mL of sterile 0.01% Tween 80 in 15 mL tubes. After vortexing and filtering, 12 μ L of spore solution was transferred to each side of a hemocytometer to determine concentration.

Analysis of Ethanol Concentration in Bolts

To estimate the ethanol environment experienced by the experimental ambrosia beetles and entomopathogenic fungi, we measured the surface (cambium) and internal (sapwood and heartwood) ethanol content. Bolts (~24 cm long and ~3.5 cm diameter; $n=12$) were prepared from American beech trees (*F. grandifolia*) and soaked in water ($n=6$) or 15% ethanol ($n=6$) for 72 h. Bolts were then held at 25°C for 24 h ($n=6$) and 7 d ($n=6$). The bolts were then transferred and stored at -80°C until processing.

Technical replicates were collected from each bolt, whereby samples were taken at 3 points along the length: 12.5 mm from each end and the middle at 25 mm. Drill samples were taken at 4 equidistant points (1.0 cm) around the circumference for each technical replicate, which included the cambium, sapwood, and heartwood tissue. The 4 drill shavings for each technical replicate drill were pooled, and 30 mg was weighed out for GC-MS analysis. Based on Parajuli et al. (2025), a 10 μ L aliquot of 0.5 μ g/mL tert-butanol-d10 representing 500 ng (Sigma-Aldrich, St. Louis, MO, USA) was pipetted onto filter paper in each sample as an internal standard. Samples were heated at 40°C for 5 min before exposure to the SPME fiber (carboxen/polydimethylsiloxane CAR/PDMS; 75 μ m coating; Sigma-Aldrich, St. Louis, MO, USA) for 5 min. The same SPME fiber was used for all headspace samples. The fiber was thermally desorbed for 2 min at 250°C in the injection port of an Agilent 7890B GC (Agilent Technologies, Palo Alto, CA, USA) with a SPME liner (0.75 mm i.d. $\times$ 6.35 mm $\times$ 78.5 mm, i.d. $\times$ o.d. $\times$ length; Restek, Bellefonte, PA, USA). A capillary nonpolar DB-5MS column (0.25 mm $\times$ 30 m $\times$ 0.25 μ m, inner diameter $\times$ length $\times$ film thickness; Agilent J&W, Santa Clara, CA, USA) was used for analysis. An Agilent 5977A mass spectrometer was operated in electron impact mode with a scan range of 33 to 120 amu. The combined peak area for ethanol using the ions m/z 45 and 46 was divided by the peak area of the internal standard tert-butanol-d10 using the ion m/z 65 for each sample analysis. Serial dilutions of ethanol were used as external standards, which were spiked with tert-butanol-d10 to prepare standard curves for determining the concentration of ethanol in the

tissue samples (99.5% purity, Acros Organics/Thermo Scientific Chemicals, Waltham, MA, USA) (Romeo 2009). See [Supplementary Materials](#) for further details.

Statistical Analyses

All analyses were conducted in R (v. 4.5.1), using RStudio (v. 2025.09.0 + 387).

Plate Bioassay

Survival curves from the entomopathogenic fungi plate bioassay were analyzed using Kaplan–Meier estimates, with survival per day as the response variable, and fungal treatment (character) and concentration (factor) as the predictors. Log-rank comparisons of Kaplan–Meier estimates were conducted using the `survdiff` function from the R package `survival` (v3.8-3). Beetle species were analyzed separately. To determine the differences between beetle species, fungal species, and fungal concentrations, we conducted post hoc pairwise tests using the `pairwise_survdiff` function from the `survminer` package (v0.5.0). Prior to visualization, data were subjected to a threshold cutoff wherein control mortality cannot be reduced below 80%. In replicates that failed to meet this threshold, data were excluded from visualization from the day prior to control mortality dropping below 80%. Days were only included in visualization if they had at least 3 replicates that met this criterion. For visualization purposes, data were corrected using Abbott's Formula ($1 - (X - Y)/X$, where X = survival of control treatment and Y = survival of experimental treatment) which corrects for natural deaths observed in the control group.

X. germanus Bolt Bioassay

Beetle mortality in response to entomopathogenic fungal species in ethanol-soaked bolts was analyzed with a generalized linear mixed effects model with a binomial distribution using the `glmer` function from the `lme4` package (v. 1.1-37). The model included treatment application (beetles vs. bolts), fungal species, and their interaction term as factorial fixed effects. Individual bolt ID was included in the model as a random intercept. Foundresses that could not be located upon bolt dissection were dropped from analyses. Mycosed and dead individuals were combined into a single “mortality” class, since fungal infections can cause insect death, then fail to present mycosis. No control beetles (beetles or bolts treated with water) exhibited mycosis.

The number of brood produced by foundresses in response to treatment application and fungal species was analyzed with a linear mixed effects model using maximum likelihood with the `lme` function from the `nlme` package (v 3.1-168). The total number of brood (eggs and larvae combined) was used as the response variable, and treatment application, fungal species, and their interaction term were included as factorial fixed effects. The individual foundress ID was included as a random intercept. Galleries that had no brood were included in the analyses (total brood count = 0), while foundresses that failed to dig and therefore produced no brood were excluded. For post-hoc analyses, we calculated the estimated marginal means using the `emmeans()` function from the `emmeans` package (v. 1.11.2).

Ethanol Concentration Bolt Bioassay

The effect of ethanol concentration in soaked wood bolts on beetle mortality when exposed to *B. bassiana* (EM044) was not statistically analyzed due to high mortality rates in all

treatments. As there was 100% mortality in all but 1 treatment, statistical models fail to converge due to complete separation.

Beauveria Strain Ethanol Tolerance Bioassay

The effect of ethanol concentration on the radial growth of 2 *B. bassiana* strains (Botanigard, EM021 and EM044) was analyzed using a linear mixed effects model with maximum likelihood using the `lme` function from the `nlme` package. Fungal colony growth (mm²) across 4 time points was used as the response variable, while ethanol content (factor), strain (factor), and their interaction term were included as fixed effects. Individual plate ID was included as a random intercept.

The physiological tradeoff between hyphal growth and spore production at 0% ethanol versus high ethanol concentrations (5%) was analyzed for both strains using a linear mixed effects model with maximum likelihood. The mean spores per mm³ (measured via hemocytometer) was used as the response variable, and ethanol concentration (factor), strain (factor), final area (mm² at 192 h; numeric), and their interaction terms were included in the model as fixed effects. Plate ID was included as a random intercept.

Results

Plate Bioassay

We showed that 3 species of commercial biocontrol fungi could successfully infect and kill invasive ambrosia beetles (*A. maiche*, *X. crassiusculus*, and *X. germanus*) by directly treating beetles with fungal spores. The survival of all 3 beetle species was significantly reduced by all 3 biocontrol fungi (Kaplan–Meier analyses; $DF = 9$ and p -value < 0.0001 for each species); all 3 species of fungi were able to kill the 3 species of beetles at high concentrations (1×10^7 to 5×10^7), with lower concentrations having differing success dependent on beetle and fungal species (Fig. 2). The efficacy and LT50 depended on both fungi and beetle species, with *A. maiche* being the most susceptible beetle species, while *X. germanus* was the least susceptible. *B. bassiana* proved the most efficient biocontrol fungi, with *C. javanica* being the least efficacious (Fig. 2, [Supplementary Table S1](#)). The highest concentrations of *B. bassiana* (EM021) and *M. brunneum* were equally lethal across all species of beetle, with LT50s of 2 to 4 d ([Supplementary Tables S1–S4](#)). The mid and low concentrations of *C. javanica* were not significantly different from the low dose of *M. brunneum* and *B. bassiana* for *X. germanus*, which was the most robust beetle species tested (Fig. 2; [Supplementary Table S3](#)).

X. germanus Bolt Bioassay

When *X. germanus* beetles were exposed to biocontrol fungi on ethanol-soaked beech bolts, we found that even in our positive controls of directly treated beetles, this exposure to ethanol decreased beetle mortality from 80% to 100% in the previous Plate Bioassay to 8% to 62%, depending on fungal species (values corrected using Abbott's Formula; Fig. 3A). When untreated beetles were released onto wood sprayed with biocontrol fungi, survival at 14 d significantly increased from the directly treated beetles for all fungal species (binomial GLMM, $\chi^2 = 40.22$, $DF = 1$, $P < 0.0001$; [Supplementary Table S5](#); Fig. 3A). The pathogenicity of each fungal species differed both within and across application treatments (Fungi: $\chi^2 = 54.21$, $DF = 3$, $P < 0.0001$; Fungi: Treatment: $\chi^2 = 18.01$,

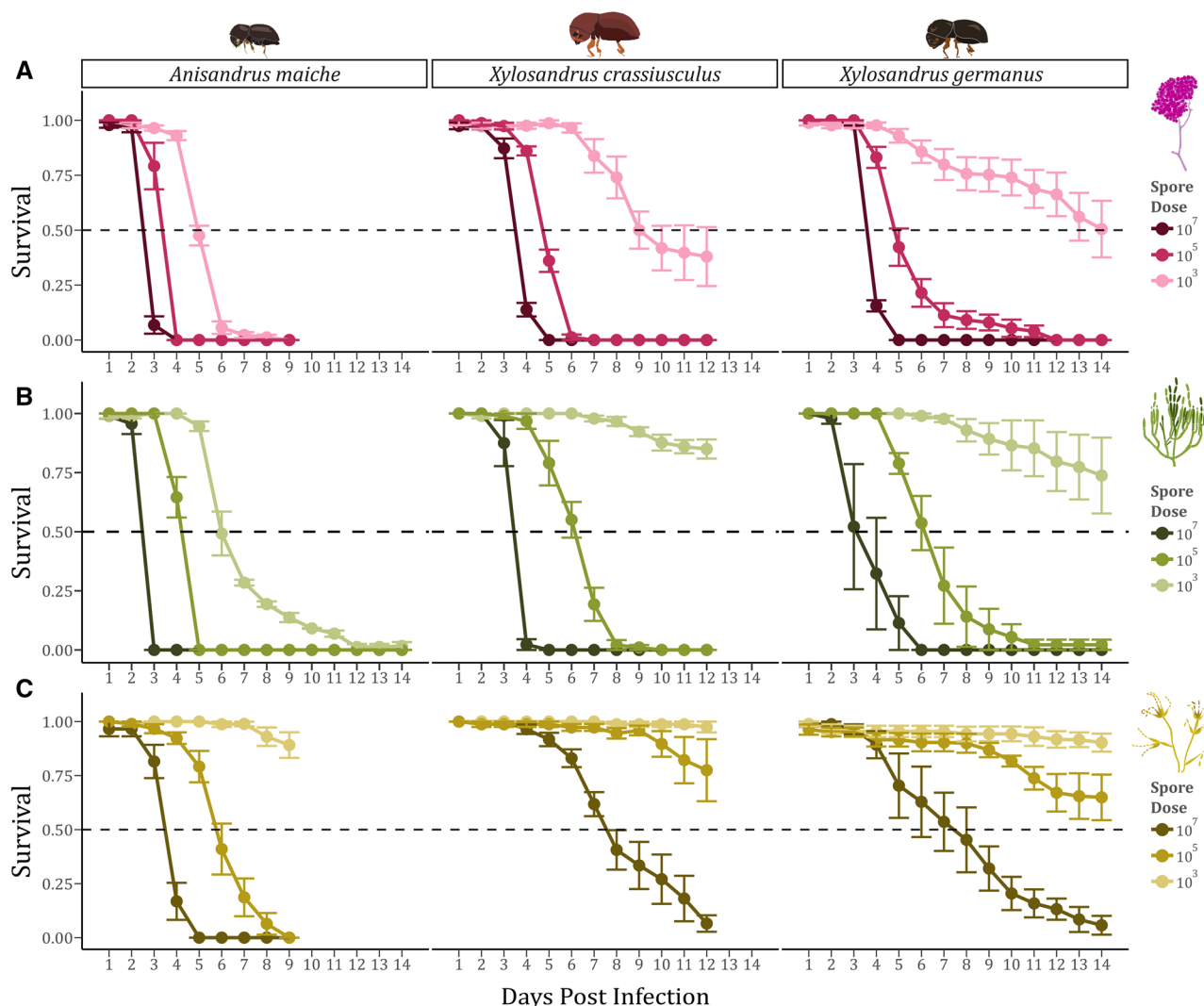


Fig. 2. The survival curves of 3 species of ambrosia beetle (*Anisandrus maiche*, *Xylosandrus crassiusculus*, *Xylosandrus germanus*) when directly treated with commercial biopesticide entomopathogenic fungi (A: Botanygard 22WP, *Beauveria bassiana*, EM021 [Bba]; B: Met52, *Metarhizium brunneum* [Mbr]; C: PFR-97, *Cordyceps javanica* [Cja]). Beetles were treated with 3 concentrations of fungal spores; packaged biopesticides (Botanygard and PFR-97) were applied at the highest recommended dose by the manufacturer (high), with lower concentrations (middle, low) serially diluted by 10^2 . *Metarhizium* spores were collected from fungal cultures, and diluted to 10^7 , 10^5 , and 10^3 for the concentration treatments. Survival curves were calculated with Abbott's formula which corrects for natural deaths in the control group ($1 - (X - Y)/X$, where X = survival of control treatment and Y = survival of experimental treatment). Data were excluded from survival curves once control survival decreased below 80% for more than 1 replicate. Error bars = SE.

DF=3, $P=0.0004$; Supplementary Table S5; Fig. 3A). *B. bassiana* (EM021) and *M. brunneum* performed best in the beetle treatments, but even their efficacy was decreased to 3% and 15% mortality in the bolt treatments, respectively. The weakest fungi in the Plate Bioassay (*C. javanica*) did not differ from controls in either the direct beetle or bolt treatments (Fig. 3A). Treatment mortality was strongly correlated with other measures of life history, such as gallery structure, *Ambrosiella* symbiont development, and brood state (Supplementary Fig. S2).

Since the majority of beetles survived the 14-d experiment, we also examined the sublethal effects of fungal treatment by examining the number of eggs and larvae that females produced (Supplementary Table S6). *B. bassiana* (EM021) significantly reduced the number of brood that surviving females produced when directly treated (post hoc estimated marginal means: DF=515, t -ratio=-4.446, $P=0.0003$;

Fig. 3B). *M. brunneum* had no significant effects on brood numbers between the beetle and bolt treatments. Despite having minimal effects on mortality, *C. javanica* significantly reduced the number of brood in directly-treated beetles compared to the bolt treatment (post hoc estimated marginal means: DF=515, t -ratio=-3.021, $P=0.0533$; Fig. 3B). However, *C. javanica* brood numbers did not significantly differ from the control group in either beetle or bolt treatments (Fig. 3B).

We analyzed the ethanol content of beech bolts soaked in 15% ethanol (the same regimen as the bolts used in the *X. germanus* Bolt Bioassay), to approximate the environmental ethanol encountered by both beetles and fungi. Wood soaked in 15% ethanol contained 55 times more ethanol than water-soaked wood at the end of 7 d, indicating that our preparation of the bolts was adequate to infuse the wood

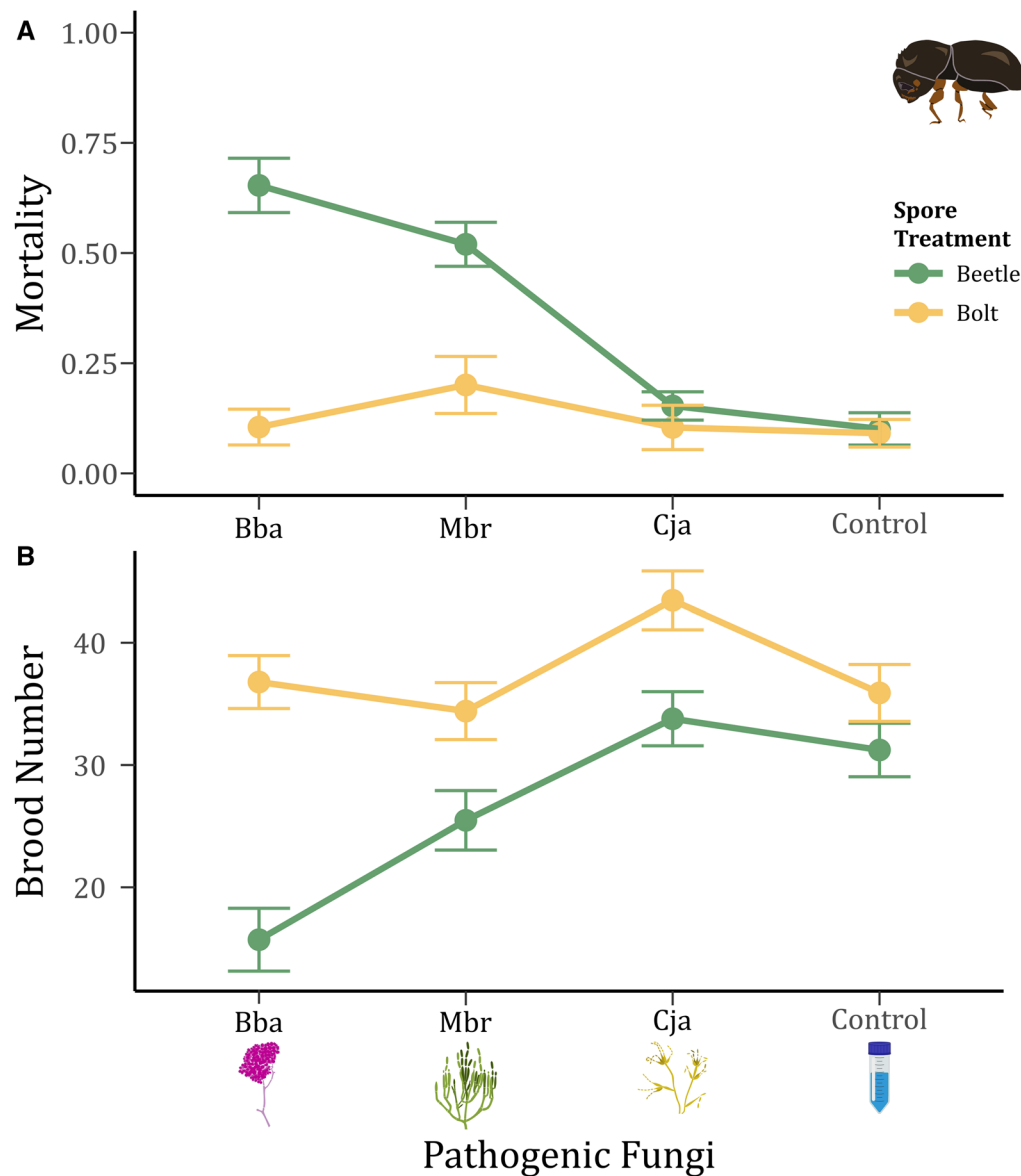


Fig. 3. Effects of commercial entomopathogenic fungi (Botanigard 22WP, *Beauveria bassiana*, EM021 [Bba]; Met52, *Metarhizium brunneum* [Mbr]; PFR-97, *Cordyceps javanica* [Cja]) on *Xylosandrus germanus* foundress survival and brood production. Beetles were either directly treated with fungal spores (green line), or placed on wood bolts after fungal spores had been applied to the surface (yellow line). Treatments consisted of 24 beetles, replicated 4 times. After 14 d the bolts were cut and galleries dissected to determine foundress survival (A) and number of brood produced (B). Error bars = SE.

with long-lasting ethanol. The ethanol content in the wood increased from the first measurement taken directly after removal from the ethanol solution, (441.07 ± 52.17 ng/mg of tissue) to the 2nd measurement, after 7 d in an environmental chamber (769.89 ± 83.63 ng/mg of tissue). Similar concentrations of ethanol have been observed in naturally flooded trees attacked by ambrosia beetles (Parajuli et al. 2023, 2025), though some studies report much lower concentrations (Ranger et al. 2013, 2018), with no clear indication of cause.

Ethanol Concentration Bolt Bioassay

We investigated how the concentration of ethanol in which wood bolts were soaked affected the pathogenicity of *B. bassiana* (EM044) against *X. germanus*. In addition to the 15%

ethanol treatment used previously, bolts were also soaked in 5% and 0% ethanol (distilled water). Beetles were all directly treated with fungal spores, and a positive control was set up on 4% water agar plates. In contrast to the results of the initial *X. germanus* Bolt Bioassay, treatments resulted in 100% of beetles dead within 4 to 5 d (other than 3% of beetles that survived in the 15% ethanol treatment), with nearly 100% mycosis rate (Supplementary Fig. S3).

Beauveria Strain Ethanol Tolerance Bioassay

We investigated if ethanol tolerance could explain the differences in pathogenicity and lethality of the 2 strains of *B. bassiana* (EM021 and EM044). Both strains were sourced from Botanigard 22WP, purchased a year apart. The Plate Bioassay and the *X. germanus* Bolt Bioassay used EM021, which

displayed reduced lethality when treated beetles were housed with ethanol-soaked wood, while EM044 was used in the Ethanol Concentration Bolt Bioassay and resulted in 100% beetle mortality regardless of ethanol treatment. We assessed ethanol tolerance by growing each strain on potato dextrose agar plates that contained 0%, 0.1%, 1%, or 5% ethanol by volume. While both strains showed a significant decrease in radial growth at 5% ethanol (LMM, $F_{3,32}=4.30$, $P=0.0117$; Supplementary Table S7), EM021 had a 350% decrease in final area compared to 0% ethanol. In contrast, EM044 had significantly smaller colonies than EM021 at all ethanol concentrations (LMM, $F_{1,12}=31.14$, $P<0.0001$; Supplementary Table S7), but its final area only decreased 170% at 5% ethanol (Fig. 4A). The intermediate treatment levels (0.1% and 1% ethanol) did not differ significantly from 0% ethanol in either strain.

Ethanol concentration also significantly reduced spore production for both strains of *B. bassiana* (LMM, $F_{1,12}=383.30$, $P<0.0001$; Supplementary Table S8). Similarly to area, EM021 produced significantly more spores than EM044 (LMM, $F_{1,12}=235.19$, $P<0.0001$; Supplementary Table S8), but also had a significantly stronger reduction in spore concentration at 5% ethanol (LMM, $F_{1,12}=6.99$, $P=0.0214$). EM021 produced 17 times fewer spores at 5% ethanol, while EM044 spore production was only reduced by half (Fig. 4B). In addition, the final area attained by each strain was significantly correlated with the number of spores produced (LMM, $F_{1,12}=256.79$, $P<0.0001$; Supplementary Table S8). At 5% ethanol, both strains had smaller colonies and lower spore concentrations; however, EM044 had significantly more spores produced from smaller colonies (LM, $F_{1,3}=14.72$, $P=0.03122$, adj. $R^2=0.7743$; Fig. 5). While our sample size

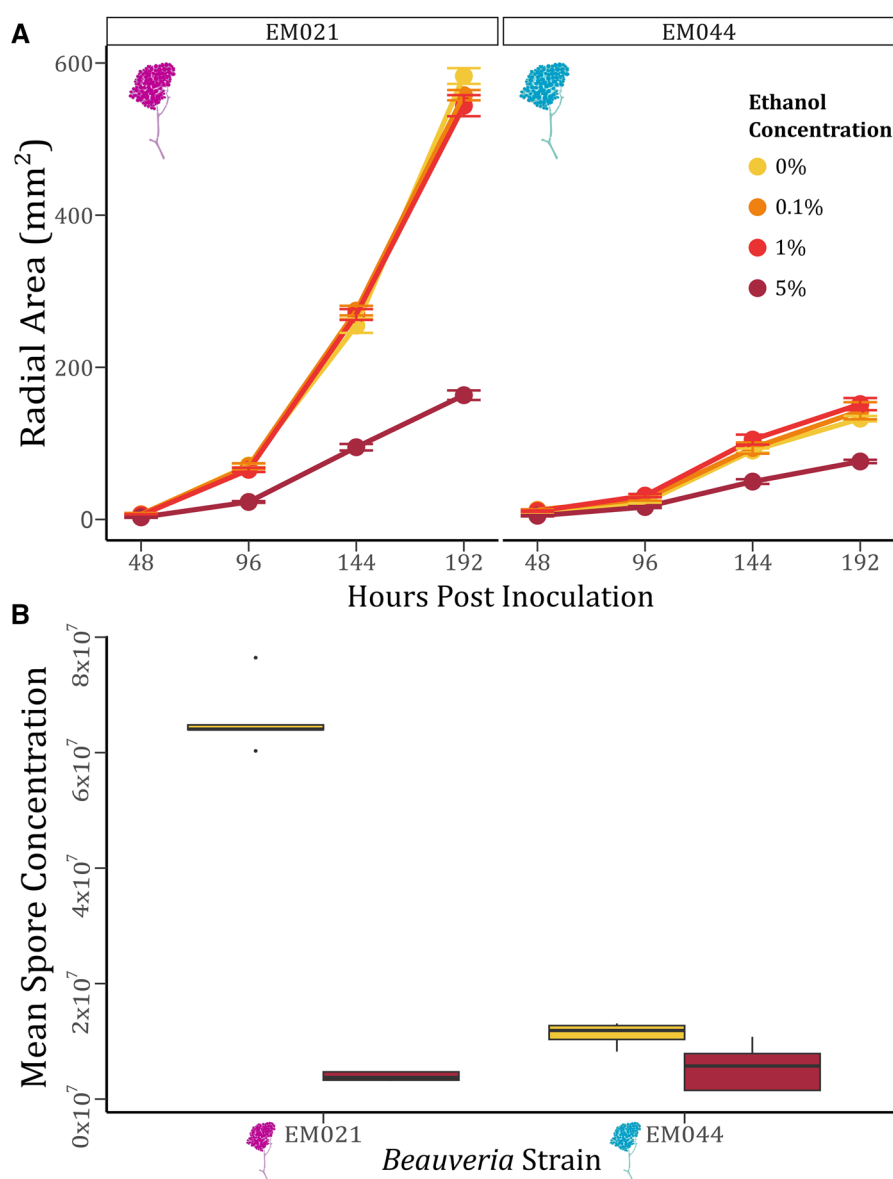


Fig. 4. Effects of ethanol concentration on the radial growth (A) and spore production (B) of 2 Botanigard 22WP *Beauveria bassiana* strains (EM021: Plate bioassay, *X. germanus* Bolt Bioassay; EM044: Ethanol Concentration Bolt Bioassay). Fungal colonies were point inoculated onto agar plates containing 4 concentrations of ethanol. Radial growth of the colonies was imaged every 48 h for 8 d, and images were measured using the area tool in Fiji. Spore concentrations were measured for 0% and 5% ethanol treatments at 192 h. Error bars = SE.

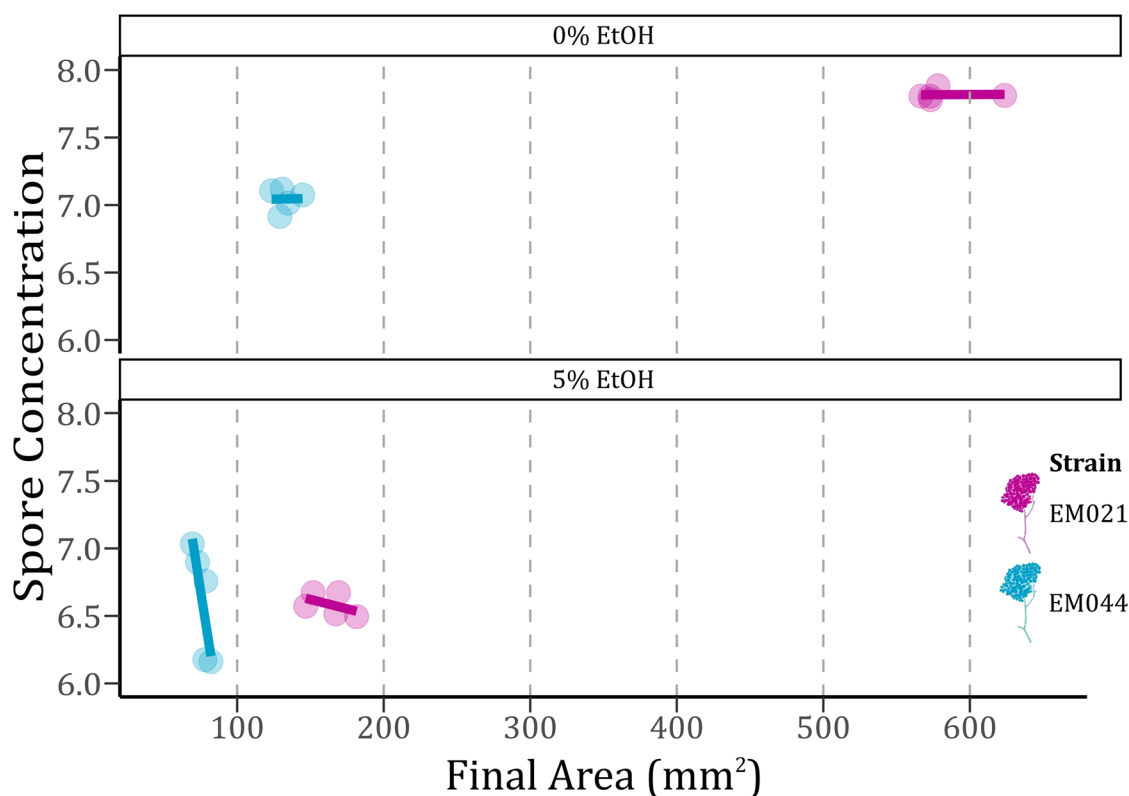


Fig. 5. Ethanol tolerance bioassay: Effects of ethanol concentration on the allocation of growth versus spore production for 2 Botanigard 22WP strains of *B. bassiana* (EM021: Plate bioassay, *X. germanus* Bolt Bioassay; EM044: Ethanol Concentration Bolt Bioassay). Spore concentration was determined by scraping the entire surface area of the fungal colony into 0.01% Tween 80, then counting spore density on a hemocytometer. Final area was measured from images of the fungal colonies after 192 h of growth, using the area tool in Fiji.

is small (5 replicates), this could indicate a tradeoff between mycelial growth and ethanol tolerance that results in higher fitness in high ethanol environments, and a possible explanation for the differences in pathogenicity between our *B. bassiana* strains.

Discussion

Entomopathogenic fungi carry the benefit and the burden of being alive. As living pesticides, fungi can choreograph a suite of enzymes and molecules to invade and kill an insect host (Lovett and St. Leger 2017). However, these fungi—importantly, their biocontrol efficacy—are subject to environmental stressors, which change dynamically throughout insect infection (Lovett and St. Leger 2015). Different fungal strains vary in their stress tolerance and responses (Feder and Hofmann 1999, Croos and Bidochka 2001, Ekesi et al. 2003, Chelico et al. 2006, Enjalbert et al. 2006, Jin et al. 2012, Borisade and Magan 2015, Lovett and St. Leger 2015). Trade-offs between stress responses and virulence are a widespread phenomenon in pathogens, including various fungus-insect systems (Borisade and Magan 2015, Ortiz-Urquiza and Keyhani 2015, Keyhani 2018, Su et al. 2021). Research into entomopathogenic fungi focus on well-known hurdles to fungal growth: UV, temperature, and desiccation tolerance (Ekesi et al. 2003, Chelico et al. 2006, Fernandes et al. 2008, Rangel et al. 2008, Fang and St. Leger 2010, Shang et al. 2012, Liao et al. 2014). To properly interpret bioassays in insects with specialized lifestyles and habitats, such as ambrosia beetles, it is important to consider other relevant environmental stressors, such as ethanol.

Our experiments investigated comparative bioassays of 3 commercial biocontrol fungi (ie. *C. javanica*, *M. brunneum*, and *B. bassiana*) against 3 apple-associated ambrosia beetle species. In agreement with previous studies (Castrillo et al. 2011, 2013, Gugliuzzo et al. 2023b), *M. brunneum* and *B. bassiana* exhibit high pathogenicity in ambrosia beetles in isolated laboratory environments. We observed pre-lethal impacts from entomopathogenic fungal infection, including reduced brood numbers in infected foundresses, resulting in generational reductions. Reduced brood numbers from entomopathogenic fungi have also been shown with other ambrosia beetles, such as *Xyleborus affinis* (Castrejón-Antonio et al. 2023), *Xylosandrus compactus* (Gugliuzzo et al. 2023b), and *Euwallacea interjectus* (Liu et al. 2025).

Previous researchers have observed *C. javanica* to be more effective against ambrosia beetles than our results. In particular, Avery et al. (2018) found that the same strain of *C. javanica* (PFR-97) displayed equal or higher pathogenicity than *B. bassiana* and *M. brunneum* when tested against *X. crassiusculus* and other *Xylosandrus* species. Native *C. javanica* (ie. not the strain presently studied) was effective against *Anisandrus dispar* in bioassays (Kushiyevev et al. 2018). However, *C. javanica* performed poorly in our study against *Xylosandrus* species, even in the absence of ethanol in the Plate Bioassay; the reason for this discrepancy is not immediately apparent. We ultimately chose not to include this fungus in our subsequent experiments: it is doubtful that a blastospore formulation of *C. javanica* PFR-97 will be effective in natural conditions against *Xylosandrus* species infesting various tree systems. Blastospores are naturally present within the relatively stable host hemolymph, not the external

environment, and so are more susceptible to environmental stressors than spores.

With our initial plate bioassay, we were able to interpret differences across beetle and fungus species. We found differences in mortality due to fungal infection across beetle species and spore concentrations. In our plate bioassays, we found that *A. maiche* was more susceptible to entomopathogenic fungi than our tested *Xylosandrus* species, which displayed similar mortality rates across fungi, with *X. germanus* being the most resilient. This was observed in *A. maiche*, both with higher mortality at lower spore concentrations and earlier mortality (ie. LT50) at similar spore concentrations. We suggest that these unique observations are related to *A. maiche* being a smaller beetle with a thinner cuticle. Fungi may penetrate a thinner cuticle faster and lead to earlier onset of infection and mortality. We observed during handling and rearing that *A. maiche* was much more likely to be damaged physically, which may be further evidence of *A. maiche* having a thinner cuticle than *Xylosandrus* species. How and whether this effect translates into the naturally high-ethanol wood environment is to be determined. Still, the evidence we have emphasizes the importance of species diagnostics for growers determining effective integrated pest management solutions for those myriad beetle pests they face.

Ethanol is a fundamental cue for ambrosia beetle habitat selection, and, more recently, ethanol tolerance and responses have been shown to be critical for the co-evolution and maintenance of ambrosial fungi (Ranger et al. 2018, Lehenberger et al. 2020, Rassati et al. 2020, Gugliuzzo et al. 2023a). In our pathosystem, we found that all of our tested fungi are less effective as pathogens in the presence of ethanol. Ethanol causes lethal osmotic and chaotropic stress in fungi (Hallsworth 1998, Hallsworth et al. 1998, Dao and Dantigny 2011, Cray et al. 2015), which underlies its widespread use as an antiseptic. Our results suggest that the ethanol-enriched environments where most ambrosia beetles select to reside may benefit beetles by protecting against infection. In support of this idea, we found increased ethanol tolerance was directly correlated with in-wood pathogenicity in closely related strains of *B. bassiana*. Our results highlight the critical role of ethanol in the ambrosia beetle system. First, flooding induces root microbial dysbiosis and anoxia, leading to anaerobic respiration that increases ethanol concentrations in tree tissues (MacDonald and Kimmerer 1991, Agarwal and Grover 2006, Ferner et al. 2012, Martínez-Arias et al. 2020, 2022). Beetles are attracted behaviorally to ethanol in these stressed tissues (Moeck 1970, Vité and Bakke 1979, Oliver and Mannion 2001, Reding et al. 2017, Ranger et al. 2021, Cavaletto et al. 2023). This high-ethanol environment has been shown to induce symbiotic traits in ambrosial fungi and limit contaminating fungi in beetle gardens, including insect pathogenic fungi (Ranger et al. 2015, 2018, Lehenberger et al. 2020). Essentially, *Ambrosiella* species have been selected by beetles to colonize microbially hostile environments (Kampf 2024).

Our studies extend the purview of ethanol benefits to include protecting beetles from fungal infections. Foundresses are the only beetles that are exposed to the external environment, and even so, usually only for a matter of hours to days. In this context, their immediate seeking of a high-ethanol environment provides anti-fungal protection during early stages of infection (ie. attachment and germination). This is when fungi would likely be most susceptible to succumb to ethanol stresses, particularly as these infection stages proceed externally. Overall,

it is clear that ethanol plays a multifaceted and essential role in ambrosia beetle colonization. Ambrosia beetle immune protection through behavioral choice may represent a kind of behavioral immunity available to insects that live in high-ethanol or other high-stress habitats. We speculate that this phenomenon may be widespread: underlying biocontrol success in diverse pests occupying a naturally ethanol-rich environment or creating 1 through symbiosis with ethanol-producing fungi, such as *Ambrosiella* spp. or yeasts associated with spotted wing *Drosophila*.

Regarding the development of beetle biocontrol agents, our studies suggest that strains with high ethanol tolerance during early developmental stages would be most effective against ambrosia beetles under field conditions. For fungi, high ethanol tolerance is likely attainable rapidly in entomopathogens through artificial selection in the laboratory. Fungal strains could also be engineered to tolerate high ethanol with the introduction of exogenous genes or overexpression of native genes, such as alcohol dehydrogenase (Lovett and St. Leger 2018). In our study, we found that our highly artificial bioassay (ie where beetles were not provided a natural environment) resulted in higher pathogenicity, but this markedly reduced as soon as beetles were allowed to access a more-natural, high-ethanol wood environment. The stark differences in ethanol tolerance seen in our 2 *B. bassiana* strains, which were isolated from the same product and manufacturer, highlight the necessity of considering strain variation in the development of biological control. Entomopathogenic fungi often present strain differences and lab-to-field differences in efficacy, which makes it difficult to appreciate their potential (Castrejón-Antonio and Tamez-Guerra 2025). State-of-the-art quality control in fungal production primarily seek to establish species-level identification, but not strain-level identification (Jenkins and Grzywacz 2000, Jaronski and Mascarín 2017). Moving forward, it may be beneficial to validate strain identity and stress tolerances when developing biological control for pests in high-stress environments. While our results highlight future challenges—and possible solutions—for biological control of ambrosia beetles, the breadth of our experimental approaches allows us to appreciate both strain-specific and habitat-specific effects of these entomopathogenic fungi in the presence of ethanol.

Our studies led us to focus on the role of ethanol in bioassays, but acetic acid has recently been implicated as an important factor for ambrosia beetle development in vitro and in vivo (Reding et al. 2026). The impact of other flooded tree products, especially those known to produce physiological stress in microbes, on biocontrol agents such as *M. brunneum* and *B. bassiana* should be investigated. Success of direct versus indirect application of fungal spores to beetles is inconsistent across studies; nonetheless, direct application in the field is impractical, in contrast to spraying tree surfaces (Castrillo et al. 2013, Carrillo et al. 2015, Kushiyevev et al. 2018, Tuncer et al. 2019, Nel et al. 2023). While preliminary studies found that surface treatment of wood delivered a low spore dose (unpubl.), efforts to formulate products to improve spore adherence may help spores overcome beetle immune protection. Experiments examining different application methods tailored to ambrosia beetles could further inform recommended application rates for biocontrol.

In our effort to evaluate the efficacy of various biocontrol fungi against ambrosia beetles, we found that strain differences and the chemical composition of the host habitat are

fundamental considerations. Commercially available microbes (ie. fungi and bacteria) that are antagonistic to other fungi are also under investigation for ambrosia beetle management, given the reliance of the beetle on its fungal symbiont (Castrillo et al. 2016, Gugliuzzo et al. 2022, Costanzo et al. 2025). These microbial pesticides result in reduced symbiont growth and brood production, indicating potential for population control of ambrosia beetles (Castrillo et al. 2016, Gugliuzzo et al. 2022, Costanzo et al. 2025). However, considering our results, the effects of ethanol on these microbial antagonists should be investigated: a key step in determining if these approaches will be successful in ecologically realistic environments of the ambrosia beetles. In this light, development of fungal strains that target specific insects—as opposed to generalist pathogen products, which have thus far found most commercial success—may prove to be the most effective strategy when dealing with specialized pests, including ambrosia beetles.

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Author Contributions

M. Elizabeth Moore (Conceptualization [equal], Data curation [lead], Formal analysis [lead], Investigation [lead], Methodology [lead], Project administration [equal], Supervision [equal], Validation [equal], Visualization [lead], Writing—original draft [equal], Writing—review & editing [equal]), Anne E. Curé (Data curation [supporting], Project administration [supporting], Resources [supporting], Supervision [supporting], Writing—review & editing [supporting]), Thomas W. Johnson (Data curation [supporting], Investigation [supporting], Methodology [supporting], Writing—review & editing [supporting]), Gregory Biddle (Data curation [supporting], Investigation [equal], Writing—review & editing [supporting]), Julie Baniszewski (Data curation [supporting], Validation [supporting], Writing—review & editing [equal]), Christopher M. Ranger (Conceptualization [supporting], Funding acquisition [lead], Methodology [supporting], Project administration [equal], Validation [equal], Writing—review & editing [supporting]), Lindsey R. Milbrath (Conceptualization [equal], Funding acquisition [equal], Investigation [equal], Project administration [supporting], Resources [supporting], Supervision [supporting], Writing—original draft [equal], Writing—review & editing [equal]), and Brian Lovett (Conceptualization [equal], Data curation [supporting], Formal analysis [equal], Methodology [supporting], Project administration [equal], Resources [lead], Supervision [equal],

Visualization [equal], Writing—original draft [equal], Writing—review & editing [equal])

Supplementary Material

Supplementary material is available at *Journal of Economic Entomology* online.

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Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data and R code is available via GitHub and Zenodo repository at <https://doi.org/10.5281/zenodo.20328132>

References

- Agarwal S, Grover A. 2006. Molecular biology, biotechnology and genomics of flooding-associated low O₂ stress response in plants. *Crit. Rev. Plant Sci.* 25:1–21. <https://doi.org/10.1080/07352680500365232>
- Agnello AM, Combs DB, Filgueiras CC, et al. 2021. Reduced infestation by *Xylosandrus germanus* (Coleoptera: Curculionidae: Scolytinae) in apple trees treated with host plant defense compounds. *J. Econ. Entomol.* 114:2162–2171. <https://doi.org/10.1093/je/toab153>
- Alabouvette C, Cordier C. (2011) Risks of microbial biocontrol agents and regulation: are they in balance? In: Ehlers R-U, editor. *Regulation of biological control agents*. Springer, p. 157–173.
- Avery PB, Bojorquez V, Gámez C, et al. 2018. Spore acquisition and survival of ambrosia beetles associated with the laurel wilt pathogen in avocados after exposure to entomopathogenic fungi. *Insects* 9:49. <https://doi.org/10.3390/insects9020049>
- Bagnères A-G, Pichon A, Hope J, et al. 2009. Contact versus feeding intoxication by fipronil in reticulitermes termites (Isoptera: Rhinotermitidae): laboratory evaluation of toxicity, uptake, clearance, and transfer among individuals. *J. Econ. Entomol.* 102:347–356. <https://doi.org/10.1603/029.102.0145>
- Bale JS, van Lenteren JC, Bigler F. 2008. Biological control and sustainable food production. *Philos. Trans. R Soc. Lond. B Biol. Sci.* 363:761–776. <https://doi.org/10.1098/rstb.2007.2182>
- Barratt BIP, Moran VC, Bigler F, et al. 2018. The status of biological control and recommendations for improving uptake for the future. *BioControl* 63:155–167. <https://doi.org/10.1007/s10526-017-9831-y>
- Bonatterra A, Badosa E, Daranas N, et al. 2022. Bacteria as biological control agents of plant diseases. *Microorganisms* 10:1759. <https://doi.org/10.3390/microorganisms10091759>
- Borisade O, Magan N. 2015. Resilience and relative virulence of strains of entomopathogenic fungi under interactions of abiotic stress. *Afr. J. Microbiol. Res.* 9:988–1000. <https://doi.org/10.5897/ajmr2015.7416>
- Bras A, Roy A, Heckel DG, et al. 2022. Pesticide resistance in arthropods: ecology matters too. *Ecol. Lett.* 25:1746–1759. <https://doi.org/10.1111/ele.14030>
- Carrillo D, Dunlap CA, Avery PB, et al. 2015. Entomopathogenic fungi as biological control agents for the vector of the laurel wilt disease,

- the redbay ambrosia beetle, *Xyleborus glabratus* (Coleoptera: Curculionidae). *Biol. Control* 81:44–50. <https://doi.org/10.1016/j.biocontrol.2014.10.009>
- Castrejón-Antonio JE, Tamez-Guerra P. 2025. Overview and recent advances in bioassays to evaluate the potential of entomopathogenic fungi against ambrosia beetles. *Insects* 16:615. <https://doi.org/10.3390/insects16060615>
- Castrejón-Antonio JE, Tamez-Guerra P, García-Ortiz N, et al. 2023. Biocontrol of *Xyleborus affinis* (Curculionidae: Scolitinae) females and progeny by *Beauveria bassiana* (Hypocreales: Cordycipitaceae) in a sawdust artificial diet model. *Insects* 14:477. <https://doi.org/10.3390/insects14050477>
- Castro LA, Griggs MH, Ranger CM, et al. 2011. Virulence of commercial strains of *Beauveria bassiana* and *Metarhizium brunneum* (Ascomycota: Hypocreales) against adult *Xylosandrus germanus* (Coleoptera: Curculionidae) and impact on brood. *Biol. Control* 58:121–126. <https://doi.org/10.1016/j.biocontrol.2011.04.010>
- Castro LA, Griggs MH, Vandenberg JD. 2013. Granulate ambrosia beetle, *Xylosandrus crassiusculus* (Coleoptera: Curculionidae), survival and brood production following exposure to entomopathogenic and mycoparasitic fungi. *Biol. Control* 67:220–226. <https://doi.org/10.1016/j.biocontrol.2013.07.015>
- Castro LA, Griggs MH, Vandenberg JD. 2016. Competition between biological control fungi and fungal symbionts of ambrosia beetles *Xylosandrus crassiusculus* and *X. germanus* (Coleoptera: Curculionidae): Mycelial interactions and impact on beetle brood production. *Biol. Control* 103:138–146.
- Cavaletto G, Ranger CM, Reding ME, et al. 2023. Species-specific effects of ethanol concentration on host colonization by four common species of ambrosia beetles. *J. Pest. Sci.* 96:833–843. <https://doi.org/10.1007/s10340-022-01537-w>
- Chelico L, Haughian JL, Khachatourians GG. 2006. Nucleotide excision repair and photoreactivation in the entomopathogenic fungi *Beauveria bassiana*, *Beauveria brongniartii*, *Beauveria nivea*, *Metarhizium anisopliae*, *Paecilomyces farinosus* and *Verticillium lecanii*. *J. Appl. Microbiol.* 100:964–972. <https://doi.org/10.1111/j.1365-2672.2006.02844.x>
- Costanzo MB, Vitale A, Biondi A, et al. 2025. Exploring the potential of synthetic and biological fungicides for managing the fungus-farming ambrosia beetle *Xylosandrus compactus*. *PLoS One*. 20:e0329063.
- Cray JA, Stevenson A, Ball P, et al. 2015. Chaotropy: a key factor in product tolerance of biofuel-producing microorganisms. *Curr. Opin. Biotechnol.* 33:228–259. <https://doi.org/10.1016/j.copbio.2015.02.010>
- Croos JNAD, Bidochka MJ. 2001. Cold-induced proteins in cold-active isolates of the insect-pathogenic fungus *Metarhizium anisopliae*. *Mycol. Res.* 105:868–873. <https://doi.org/10.1017/S0953756201004099>
- Cruz LP, Gaitan AL, Gongora CE. 2006. Exploiting the genetic diversity of *Beauveria bassiana* for improving the biological control of the coffee berry borer through the use of strain mixtures. *Appl. Microbiol. Biotechnol.* 71:918–926. <https://doi.org/10.1007/s00253-005-0218-0>
- Dao T, Dantigny P. 2011. Control of food spoilage fungi by ethanol. *Food Control* 22:360–368. <https://doi.org/10.1016/j.foodcont.2010.09.019>
- Ekesi S, Maniania NK, Lux SA. 2003. Effect of soil temperature and moisture on survival and infectivity of *Metarhizium anisopliae* to four tephritid fruit fly puparia. *J. Invertebr. Pathol.* 83:157–167. [https://doi.org/10.1016/S0022-2011\(03\)00069-7](https://doi.org/10.1016/S0022-2011(03)00069-7)
- Enjalbert B, Smith DA, Cornell MJ, et al. 2006. Role of the Hog1 stress-activated protein kinase in the global transcriptional response to stress in the fungal pathogen *Candida albicans*. *Mol. Biol. Cell.* 17:1018–1032. <https://doi.org/10.1091/mbc.e05-06-0501>
- Eskalen A, Stouthamer R, Lynch SC, et al. 2013. Host range of fusarium dieback and its ambrosia beetle (Coleoptera: Scolytinae) vector in Southern California. *Plant Dis.* 97:938–951. <https://doi.org/10.1094/PDIS-11-12-1026-RE>
- Fang W, St. Leger RJ. 2010. RNA binding proteins mediate the ability of a fungus to adapt to the cold. *Environ. Microbiol.* 12:810–820. <https://doi.org/10.1111/j.1462-2920.2009.02127.x>
- Feder ME, Hofmann GE. 1999. Heat-shock proteins, molecular chaperones, and the stress response: evolutionary and ecological physiology. *Annu. Rev. Physiol.* 61:243–282. <https://doi.org/10.1146/annurev.physiol.61.1.243>
- Fernandes ÉKK, Bittencourt VREP, Roberts DW. 2012. Perspectives on the potential of entomopathogenic fungi in biological control of ticks. *Exp. Parasitol.* 130:300–305. <https://doi.org/10.1016/j.exppara.2011.11.004>
- Fernandes ÉKK, Rangel DEN, Moraes ÁML, et al. 2008. Cold activity of *Beauveria* and *Metarhizium*, and thermotolerance of *Beauveria*. *J. Invertebr. Pathol.* 98:69–78. <https://doi.org/10.1016/j.jip.2007.10.011>
- Ferner E, Rennenberg H, Kreuzwieser J. 2012. Effect of flooding on C metabolism of flood-tolerant (*Quercus robur*) and non-tolerant (*Fagus sylvatica*) tree species. *Tree Physiol.* 32:135–145. <https://doi.org/10.1093/treephys/tps009>
- Fraedrich SW, Harrington TC, Rabaglia RJ, et al. 2008. A fungal symbiont of the redbay ambrosia beetle causes a lethal wilt in redbay and other lauraceae in the southeastern United States. *Plant Dis.* 92:215–224. <https://doi.org/10.1094/PDIS-92-2-0215>
- Gao L, Qiao H, Wei P, et al. 2022. Xenobiotic responses in insects. *Arch. Insect Biochem. Physiol.* 109:e21869. <https://doi.org/10.1002/arch.21869>
- Gerolt P. 1969. Mode of entry of contact insecticides. *J. Insect Physiol.* 15:563–580. [https://doi.org/10.1016/0022-1910\(69\)90255-8](https://doi.org/10.1016/0022-1910(69)90255-8)
- Goettel MS, Hajek AE, Siegel JP, et al. 2001. Safety of fungal biocontrol agents. In: Butt TM, Jackson C, Magan N, editors. *Fungi as biocontrol agents: progress, problems and potential*. p. 347–375.
- Gomez DF, Rabaglia RJ, Fairbanks KEO, et al. 2018. North American Xyleborini north of Mexico: a review and key to genera and species (Coleoptera, Curculionidae, Scolytinae). *Zookeys*. 768:19–68. <https://doi.org/10.3897/zookeys.768.24697>
- Govindaraju R, Joseph SV. 2025. Comparing the effects of verbenone, methyl salicylate, and permethrin on attacks of ambrosia beetles (Coleoptera: Curculionidae Scolytinae) in ornamental nurseries. *J. Insect Sci.* 25:ieaf008. <https://doi.org/10.1093/jisesa/ieaf008>
- Gugliuzzo A, Aiello D, Biondi A, et al. 2022. Microbial mutualism suppression by *Trichoderma* and *Bacillus* species for controlling the invasive ambrosia beetle *Xylosandrus compactus*. *Biol. Control* 170:104929. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0329063>
- Gugliuzzo A, Biedermann PHW, Carrillo D, et al. 2021. Recent advances toward the sustainable management of invasive *Xylosandrus ambrosia* beetles. *J. Pest. Sci.* 94:615–637. <https://doi.org/10.1007/s10340-021-01382-3>
- Gugliuzzo A, Giuliano G, Rizzo R, et al. 2023a. Lethal and sublethal effects of synthetic and bioinsecticides toward the invasive ambrosia beetle *Xylosandrus compactus*. *Pest Manag. Sci.* 79:1840–1850.
- Gugliuzzo A, Kreuzwieser J, Ranger CM, et al. 2023b. Volatiles of fungal cultivars act as cues for host-selection in the fungus-farming ambrosia beetle *Xylosandrus germanus*. *Front. Microbiol.* 14:1151078. <https://doi.org/10.3389/fmicb.2023.1151078>
- Hallsworth JE. 1998. Ethanol-induced water stress in yeast. *J. Ferment. Bioeng.* 85:125–137. [https://doi.org/10.1016/S0922-338X\(97\)86756-6](https://doi.org/10.1016/S0922-338X(97)86756-6)
- Hallsworth JE, Nomura Y, Iwahara M. 1998. Ethanol-induced water stress and fungal growth. *J. Ferment. Bioeng.* 86:451–456. [https://doi.org/10.1016/S0922-338X\(98\)80150-5](https://doi.org/10.1016/S0922-338X(98)80150-5)
- Hallsworth JE, Prior BA, Nomura Y, et al. 2003. Compatible solutes protect against chaotrope (ethanol)-induced, nonosmotic water stress. *Appl. Environ. Microbiol.* 69:7032–7034. <https://doi.org/10.1128/AEM.69.12.7032-7034.2003>
- Harrington TC, McNew D, Mayers C, et al. 2014. *Ambrosiella roeperi* sp. nov. is the mycetangial symbiont of the granulate ambrosia beetle, *Xylosandrus crassiusculus*. *Mycologia* 106:835–845. <https://doi.org/10.3852/13-354>

- Huang X, Zhao J, Zhou X, et al. 2019. How green alternatives to chemical pesticides are environmentally friendly and more efficient. *Eur. J. Soil Sci.* 70:518–529. <https://doi.org/10.1111/ejss.12755>
- Humber RA. 2008. Evolution of entomopathogenicity in fungi. *J. Invertebr. Pathol.* 98:262–266. <https://doi.org/10.1016/j.jip.2008.02.017>
- Jaronski ST, Mascarin GM. 2017. Chapter 9—Mass production of fungal entomopathogens. In: Lacey LA. editor. *Microbial control of insect and mite pests*. Academic Press, p. 141–155.
- Jenkins NE, Grzywacz D. 2000. Quality control of fungal and viral biocontrol agents - assurance of product performance. *Biocontrol Sci. Technol.* 10:753–777. <https://doi.org/10.1080/09583150020011717>
- Jin K, Ming Y, Xia YX. 2012. MaHog1, a Hog1-type mitogen-activated protein kinase gene, contributes to stress tolerance and virulence of the entomopathogenic fungus *Metarhizium acridum*. *Microbiology (Reading)* 158:2987–2996. <https://doi.org/10.1099/mic.0.059469-0>
- Kampf G. 2024. Ethanol. In: Kampf G. editor. *Antiseptic stewardship: biocide resistance and clinical implications*. Springer International Publishing, p. 23–74.
- Keyhani NO. 2018. Lipid biology in fungal stress and virulence: entomopathogenic fungi. *Fungal Biol.* 122:420–429. <https://doi.org/10.1016/j.funbio.2017.07.003>
- Kimmerer TW, MacDonald RC. 1987. Acetaldehyde and ethanol biosynthesis in leaves of plants 1. *Plant Physiol.* 84:1204–1209. <https://doi.org/10.1104/pp.84.4.1204>
- Koch E, Ole Becker J, Berg G, et al. 2018. Biocontrol of plant diseases is not an unsafe technology!. *J. Plant Dis. Prot.* 125:121–125. <https://doi.org/10.1007/s41348-018-0158-4>
- Kushiyev R, Tuncer C, Erper I, et al. 2018. Efficacy of native entomopathogenic fungus, *Isaria fumosorosea*, against bark and ambrosia beetles, *Anisandrus dispar* Fabricius and *Xylosandrus germanus* Blandford (Coleoptera: Curculionidae: Scolytinae). *Egypt J. Biol. Pest. Control* 28:55. <https://doi.org/10.1186/s41938-018-0062-z>
- Lacey LA, Grzywacz D, Shapiro-Ilan DI, et al. 2015. Insect pathogens as biological control agents: back to the future. *J. Invertebr. Pathol.* 132:1–41. <https://doi.org/10.1016/j.jip.2015.07.009>
- Lartigue S, Yalaoui M, Belliard J, et al. 2022. Consistent variations in personality traits and their potential for genetic improvement in bio-control agents: *Trichogramma evanescens* as a case study. *Evol. Appl.* 15:1565–1579. <https://doi.org/10.1111/eva.13329>
- Lehenberger M, Benkert M, Biedermann PHW. 2020. Ethanol-enriched substrate facilitates ambrosia beetle fungi, but inhibits their pathogens and fungal symbionts of bark beetles. *Front. Microbiol.* 11:590111. <https://doi.org/10.3389/fmicb.2020.590111>
- Liao X, Lu H-L, Fang W, et al. 2014. Overexpression of a *Metarhizium robertsii* HSP25 gene increases thermotolerance and survival in soil. *Appl. Microbiol. Biotechnol.* 98:777–783. <https://doi.org/10.1007/s00253-013-5360-5>
- Liu J, Jiang N, Gao H, et al. 2025. Laboratory exploration of several potential biocontrol methods against the ambrosia beetle, *Euwallacea interjectus*. *Insects* 16:56. <https://doi.org/10.3390/insects16010056>
- Lommen STE, de Jong PW, Pannebakker BA. 2017. It is time to bridge the gap between exploring and exploiting: prospects for utilizing intraspecific genetic variation to optimize arthropods for augmentative pest control—A review. *Entomol. Exp. Appl.* 162:108–123. <https://doi.org/10.1111/eea.12510>
- Lovett B, St. Leger RJ. 2015. Stress is the rule rather than the exception for *Metarhizium*. *Curr. Genet.* 61:253–261. <https://doi.org/10.1007/s00294-014-0447-9>
- Lovett B, St. Leger RJ. 2017. The insect pathogens. In: Heitman J, Howlett BJ, Crous PW, Stukenbrock EH, James TY, Gow NAR, editors. *The fungal kingdom*. John Wiley & Sons, Ltd, p. 923–943.
- Lovett B, St. Leger RJ. 2018. Genetically engineering better fungal biopesticides. *Pest Manag. Sci.* 74:781–789. <https://doi.org/10.1002/ps.4734>
- MacDonald RC, Kimmerer TW. 1991. Ethanol in the stems of trees. *Physiol. Plant* 82:582–588. <https://doi.org/10.1111/j.1399-3054.1991.tb02950.x>
- Mantzoukas S, Eliopoulos PA. 2020. Endophytic entomopathogenic fungi: a valuable biological control tool against plant pests. *Appl. Sci.* 10:360. <https://doi.org/10.3390/app10010360>
- Martínez-Arias C, Sobrino-Plata J, Macaya-Sanz D, et al. 2020. Changes in plant function and root microbiome caused by flood and drought in a riparian tree. *Tree Physiol.* 40:886–903. <https://doi.org/10.1093/treephys/tpaa031>
- Martínez-Arias C, Witzell J, Solla A, et al. 2022. Beneficial and pathogenic plant-microbe interactions during flooding stress. *Plant. Cell Environ.* 45:2875–2897. <https://doi.org/10.1111/pce.14403>
- Mascarin GM, Jaronski ST. 2016. The production and uses of *Beauveria bassiana* as a microbial insecticide. *World J. Microbiol. Biotechnol.* 32:177. <https://doi.org/10.1007/s11274-016-2131-3>
- Mayers CG, Harrington TC, Ranger CM. 2017. First report of a sexual state in an ambrosia fungus: *Ambrosiella cleistominuta* sp. nov. associated with the ambrosia beetle *Anisandrus maiche*. *Botany* 95:503–512. <https://doi.org/10.1139/cjb-2016-0297>
- Mayers CG, McNew DL, Harrington TC, et al. 2015. Three genera in the *Ceratocystidaceae* are the respective symbionts of three independent lineages of ambrosia beetles with large, complex mycetangia. *Fungal Biol.* 119:1075–1092. <https://doi.org/10.1016/j.funbio.2015.08.002>
- Mhatre PH, Divya K I, Venkatasalam E p, et al. 2022. Management of potato cyst nematodes with special focus on biological control and trap cropping strategies. *Pest Manag. Sci.* 78:3746–3759. <https://doi.org/10.1002/ps.7022>
- Moock HA. 1970. Ethanol as the primary attractant for the ambrosia beetle trypodendron lineatum (Coleoptera: Scolytidae). *Can. Entomol.* 102:985–995. <https://doi.org/10.4039/Ent102985-8>
- Monterrosa A, Joseph SV, Blaauw B, et al. 2022. Ambrosia beetle occurrence and phenology of *Xylosandrus* spp. (Coleoptera: Curculionidae: Scolytinae) in ornamental nurseries, tree fruit, and pecan orchards in Georgia. *Environ. Entomol.* 51:998–1009. <https://doi.org/10.1093/ee/nvac064>
- Muirhead KA, Murphy NP, Sallam N, et al. 2012. Phylogenetics and genetic diversity of the *Cotesia flavipes* complex of parasitoid wasps (Hymenoptera: Braconidae), biological control agents of lepidopteran stem borers. *Mol. Phylogenet. Evol.* 63:904–914. <https://doi.org/10.1016/j.ympev.2012.03.003>
- Na F, Carrillo JD, Mayorquin JS, et al. 2018. Two novel fungal symbionts *Fusarium kuroshium* sp. nov. and *Graphium kuroshium* sp. nov. of Kuroshio shot hole borer (*Euwallacea* sp. nr. *forficatus*) cause fusarium dieback on woody host species in California. *Plant Dis.* 102:1154–1164. <https://doi.org/10.1094/PDIS-07-17-1042-RE>
- Nel WJ, Slippers B, Wingfield MJ, et al. 2023. Efficacy of commercially available entomopathogenic agents against the polyphagous shot hole borer in South Africa. *Insects* 14:361. <https://doi.org/10.3390/insects14040361>
- Oliver JB, Mannion CM. 2001. Ambrosia beetle (Coleoptera: Scolytidae) species attacking chestnut and captured in ethanol-baited traps in middle Tennessee. *Environ. Entomol.* 30:909–918. <https://doi.org/10.1603/0046-225X-30.5.909>
- Ortiz-Urquiza A, Keyhani NO. 2015. Stress response signaling and virulence: insights from entomopathogenic fungi. *Curr. Genet.* 61:239–249. <https://doi.org/10.1007/s00294-014-0439-9>
- Parajuli M, Oksel C, Neupane K, et al. 2023. Acibenzolar-S-methyl induces resistance against ambrosia beetle attacks in dogwoods exposed to simulated flood stress. *J. Insect Sci.* 23:12. <https://doi.org/10.1093/jisesa/iead068>
- Parajuli M, Oksel C, Neupane K, et al. 2025. *Xylosandrus ambrosia* beetles preference of nursery tree species for attacks and colonization under water stress. *J. Insect Sci.* 25:7.
- Pell JK, Hannam JJ, Steinkraus DC. 2010. Conservation biological control using fungal entomopathogens. *BioControl* 55:187–198. <https://doi.org/10.1007/s10526-009-9245-6>

- Rabaglia RJ, Cognato AI, Hoebeke ER, et al. 2019. Early detection and rapid response: a 10-year summary of the USDA forest service program of surveillance for non-native bark and ambrosia beetles. *Am. Entomol.* 65:29–42. <https://doi.org/10.1093/ae/tmz015>
- Rabaglia RJ, Vandenberg NJ, Acciavatti RE. 2009. First records of *Anisandrus maiche* Stark (Coleoptera: Curculionidae: Scolytinae) from North America. *Zootaxa* 2137:23–28. <https://doi.org/10.5281/zenodo.188523>
- Rangel DEN, Anderson AJ, Roberts DW. 2008. Evaluating physical and nutritional stress during mycelial growth as inducers of tolerance to heat and UV-B radiation in *Metarhizium anisopliae* conidia. *Mycol. Res.* 112:1362–1372. <https://doi.org/10.1016/j.mycres.2008.04.013>
- Ranger CM, Biedermann PHW, Phuntumart V, et al. 2018. Symbiont selection via alcohol benefits fungus farming by ambrosia beetles. *Proc. Natl. Acad. Sci. USA.* 115:4447–4452. <https://doi.org/10.1073/pnas.1716852115>
- Ranger CM, Reding ME, Adesso K, et al. 2021. Semiochemical-mediated host selection by *Xylosandrus* spp. ambrosia beetles (Coleoptera: Curculionidae) attacking horticultural tree crops: a review of basic and applied science. *Can. Entomol.* 153:103–120. <https://doi.org/10.4039/tce.2020.51>
- Ranger CM, Reding ME, Schultz PB, et al. 2013. Influence of flood-stress on ambrosia beetle host-selection and implications for their management in a changing climate. *Agric. For. Entomol.* 15:56–64. <https://doi.org/10.1111/j.1461-9563.2012.00591.x>
- Ranger CM, Reding ME, Schultz PB, et al. 2016. Biology, ecology, and management of nonnative ambrosia beetles (Coleoptera: Curculionidae: Scolytinae) in ornamental plant nurseries. *J. Integr. Pest Manag.* 7(1): 9; 1–23. <https://doi.org/10.1093/jipm/pmw005>
- Ranger CM, Schultz PB, Frank SD, et al. 2015. Non-native ambrosia beetles as opportunistic exploiters of living but weakened trees. *PLoS One.* 10:e0131496. <https://doi.org/10.1371/journal.pone.0131496>
- Rassati D, Contarini M, Ranger CM, et al. 2020. Fungal pathogen and ethanol affect host selection and colonization success in ambrosia beetles. *Agric. For. Entomol.* 22:1–9. <https://doi.org/10.1111/afe.12351>
- Reding ME, Ranger CM, Baniszewski JA. 2026. Acetic acid enhances attraction to and colonization of wood bolts by ambrosia beetles (Coleoptera: Curculionidae: Scolytinae). *J. Econ. Entomol.* 119:1198–1206. <https://doi.org/10.1093/jeet/toaf343>
- Reding ME, Ranger CM, Oliver JB, et al. 2017. Ethanol-injection induces attacks by ambrosia beetles (Coleoptera: Curculionidae: Scolytinae) on a variety of tree species. *Agric. For. Entomol.* 19:34–41. <https://doi.org/10.1111/afe.12178>
- Reverchon F, Contreras-Ramos SM, Eskalen A, et al. 2021. Microbial biocontrol strategies for ambrosia beetles and their associated phytopathogenic fungi. *Front. Sustain. Food Syst.* 5:737977. <https://doi.org/10.3389/fsufs.2021.737977>
- Rim K, Price SJ, Wenninger EJ, et al. 2019. Biology and management of clover root curculio (Coleoptera: Curculionidae). *J. Integr. Pest Manag.* 10:23. <https://doi.org/10.1093/jipm/pmz020>
- Romeo JT. 2009. New SPME guidelines. *J. Chem. Ecol.* 35:1383. <https://doi.org/10.1007/s10886-009-9733-2>
- Saha N, Sharma A, Bora P. 2025. Expanding the functional landscape of microbial entomopathogens in agriculture beyond pest management. *Folia Microbiol.* 70:343–357. <https://doi.org/10.1007/s12223-025-01251-x>
- Shang Y, Duan Z, Huang W, et al. 2012. Improving UV resistance and virulence of *Beauveria bassiana* by genetic engineering with an exogenous tyrosinase gene. *J. Invertebr. Pathol.* 109:105–109. <https://doi.org/10.1016/j.jip.2011.10.004>
- Shields EJ, Testa AM. 2017. Biological control of alfalfa snout beetle with persistent entomopathogenic nematodes: expanding a single farm's success to an area-wide biological control program. *Am. Entomol.* 63:216–223. <https://doi.org/10.1093/ae/tmx046>
- Stone LBL, Bidochka MJ. 2020. The multifunctional lifestyles of *Metarhizium*: evolution and applications. *Appl. Microbiol. Biotechnol.* 104:9935–9945. <https://doi.org/10.1007/s00253-020-10968-3>
- Su X, Yan X, Chen X, et al. 2021. Calcofluor white hypersensitive proteins contribute to stress tolerance and pathogenicity in entomopathogenic fungus, *Metarhizium acridum*. *Pest Manag. Sci.* 77:1915–1924. <https://doi.org/10.1002/ps.6218>
- Thurston GS, Slater A, Nei I, et al. 2022. New Canadian and provincial records of coleoptera resulting from annual Canadian food inspection agency surveillance for detection of non-native, potentially invasive forest insects. *Insects* 13:708. <https://doi.org/10.3390/insects13080708>
- Tobin KN, Ethington MW, Ginzel MD. 2024. Volatiles from nutritional fungal symbiont influence the attraction of *Anisandrus maiche* (Coleoptera: Curculionidae) to ethanol-baited traps. *Environ. Entomol.* 53:108–115. <https://doi.org/10.1093/ee/nvad121>
- Tobin KN, Ginzel MD. 2023. The ambrosia beetle *Anisandrus maiche* (Stark) is repelled by conophthorin and verbenone and attracted to ethanol in a dose-dependent manner. *Agric. For. Entomol.* 25:103–110. <https://doi.org/10.1111/afe.12534>
- Tuncer C, Kushiye R, Erper I, et al. 2019. Efficacy of native isolates of *Metarhizium anisopliae* and *Beauveria bassiana* against the invasive ambrosia beetle, *Xylosandrus germanus* Blandford (Coleoptera: Curculionidae: Scolytinae). *Egypt. J. Biol. Pest Control* 29:28. <https://doi.org/10.1186/s41938-019-0132-x>
- Valero-Jiménez CA, Wieggers H, Zwaan BJ, et al. 2016. Genes involved in virulence of the entomopathogenic fungus *Beauveria bassiana*. *J. Invertebr. Pathol.* 133:41–49. <https://doi.org/10.1016/j.jip.2015.11.011>
- Velivelli SLS, Vos PD, Kromann P, et al. 2014. Biological control agents: from field to market, problems, and challenges. *Trends Biotechnol.* 32:493–496. <https://doi.org/10.1016/j.tibtech.2014.07.002>
- Vité JP, Bakke A. 1979. Synergism between chemical and physical stimuli in host colonization by an ambrosia beetle. *Naturwissenschaften* 66:528–529. <https://doi.org/10.1007/BF00404874>
- Wang Z-Z, Liu Y-Q, Shi M, et al. 2019. Parasitoid wasps as effective biological control agents. *J. Integr. Agricult.* 18:705–715. [https://doi.org/10.1016/S2095-3119\(18\)62078-7](https://doi.org/10.1016/S2095-3119(18)62078-7)
- Weber BC, McPherson JE. 1983. Life history of the ambrosia beetle *Xylosandrus germanus* (Coleoptera: Scolytidae). *Ann. Entomol. Soc. Am.* 76:455–462. <https://doi.org/10.1093/aesa/76.3.455>
- Young RG, Milián-García Y, Yu J, et al. 2021. Biosurveillance for invasive insect pest species using an environmental DNA metabarcoding approach and a high salt trap collection fluid. *Ecol. Evol.* 11:1558–1569. <https://doi.org/10.1002/ece3.7113>
- Zimmermann G. 2007. Review on safety of the entomopathogenic fungus *Metarhizium anisopliae*. *Biocontrol Sci. Technol.* 17:879–920. <https://doi.org/10.1080/09583150701593963>
- Zimmermann G. 2008. The entomopathogenic fungi *Isaria farinosa* (formerly *Paecilomyces farinosus*) and the *Isaria fumosorosea* species complex (formerly *Paecilomyces fumosoroseus*): biology, ecology and use in biological control. *Biocontrol Sci. Technol.* 18:865–901. <https://doi.org/10.1080/09583150802471812>